Experiment - 12 (b)

Implement a chat program using socket programming

Aim :

To implement a simple chart client server application using TCP sockets for communication between clients through a server

Algorithm :

- Start the TCP server and create a serversocket
- Accept the client connection.
- create a thread to handle each client
- continue chatting until \quit is entered
- close the connection

code :

Chatserver:

```
import java. io. *;
import java. net. *;
public class chatserver {
  static set <printwriter> clients = new Hashset
                                              <>();
    public static void main (string [] args) throws
                              exception {
      server socket server = new server socket (5000);
   system. out. println ("chat server started");
    while ( true)
    {
      socket = server. accept ();
      new thread () -{
      try {
      new inputstrem Reader [s. get Input stream ())]
      printwriter out = new printwriter()
      string msg ;
```

output

```
Server:
chat server started
After Joined the chart
Prathiksha Joined the class
  Alice : Hello
  Prathiksha : Hi

client:
   Enter your name
     Alice
     Hello Hp
      Prathiksha : Hi
```

```
  }
} your (Exceptions e)h
  start
   {
 }
 }
}
```

chat client

```
import java.io;
import java.net.*.
public  class chatclient{
  public socket = new socket ( ).True);
  s.getoutput stream (), true)
  new Thread () — h
   try thread {
       say :
   } cat ( Exception){
  }). start ();
  String msg:
  while (msg = kb-readLine())
    out.println (msg
                  equal signore ("/quit
     my;spartic
    break;

   } close() }
 }
}
```

Result:

Thus the chat client - server application was successfully implemented using TCP socket

8/10